

Se Min Kong

SOFTWARE DEVELOPER

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PROFILE

Software engineering graduate with experience connecting computer vision, ROS 2, backend services, and hardware integration to build working systems. I enjoy learning new technologies and turning ideas into systems that I can build and test.

EDUCATION

Soongsil University | College of IT, School of Software Mar 2020 - Feb 2026
Bachelor of Engineering in Software; Major in Artificial Intelligence & Big Data; Convergence Major in Big Data
GPA 3.61/4.50 | Graduated 20 Feb 2026
Dulwich College Suzhou (DCSZ) | Primary & Middle School, Suzhou, China 2008 - 2012

TECHNICAL SKILLS

Languages & Core: Python, C++, TypeScript
AI & Vision: PyTorch, Ultralytics YOLO, MediaPipe Hands, OpenCV, Transformers, CUDA
Robotics & Systems: ROS 2, DYNAMIXEL XL330, Jetson, Raspberry Pi 5, Intel RealSense, Dobot, Isaac Sim / Isaac Lab
Backend & Tools: FastAPI, Django, WebSocket, React, Node.js, Socket.io, Git, Docker, Linux (Ubuntu)

EXPERIENCE

Samsung SW·AI Academy For Youth (SSAFY) | Trainee, Robotics Track Jan 2026 - Present

- Completing an intensive SW·AI curriculum centered on robotics, computer vision, ROS 2, and integrated hardware-software projects.

Republic of Korea Army | Tactical Communications Equipment Operator/Maintainer May 2021 - Nov 2022

- Completed mandatory military service and was discharged as a Sergeant.

SELECTED PROJECTS

THING Human-Mimetic Robot Hand | Team Project, Robotics & Edge AI Jul 2026 - Aug 2026
ROS 2 | MediaPipe | OpenCV | Jetson | Raspberry Pi 5 | DYNAMIXEL | React | Django | AWS EC2 | [Case Study](#)

- Team project mapping 21 MediaPipe hand landmarks to a seven-axis HandCommand.
- Handled Blender modeling, 3D printing, synchronized seven-motor control and U2D2 communication, ROS bridge and Cyclone DDS configuration, and tendon mechanism integration.

AQIS for Smart Factory | Team Lead, Full-Stack & Robot Integration May 2026 - Jun 2026
ROS 2 | FastAPI | React | RealSense | Dobot | YOLOv5 | RoboDK | [Repository](#)

- Team project spanning physical inspection and Dobot sorting, RoboDK process simulation, and SLAM-based robot monitoring.
- Trained YOLO; configured the integrated ROS environment, RealOps and separate MJPEG video streams, package-based navigation to designated goals, and RoboDK command/state relay.

Brain Tumor MRI Classification & Segmentation | Personal Project Feb 2026 - Apr 2026
Python | PyTorch | Ultralytics YOLO11 | OpenCV | NumPy | [Repository](#)

- Implemented YOLO11 classification and segmentation, mask-to-polygon preprocessing, and a unified inference interface.
- Performance verification with independent evaluation data and execution logs remains to be completed.

Alkkagi.io | Personal Project Mar 2026 - Apr 2026
React 19 | TypeScript | Node.js | Express | Socket.io | [Repository](#)

- Built a real-time multiplayer game with React and Socket.io.
- Implemented server-authoritative state, input limits, and collision, friction, and position-correction calculations.

Briefit | Team AI Engineer, AI-Based News Curation May 2025 - Sep 2025
Python | aiohttp | BeautifulSoup | Transformers | KoBART | [Repository](#)

- Team news curation service for browsing articles from multiple outlets and reading summaries. Collected and cleaned about 4,000 articles with Crawl4AI and fine-tuned KoBART on about 3,000.
- Compared T5 and KoBART with ROUGE and integrated the Gemini API to improve service summaries.

AWARDS & CERTIFICATIONS

SSAFY Common Project - Excellence Award | AIoT Track, Gwangju Class 1 (1st Place) 10 Aug 2026
2025 IT Project Pro League - Encouragement Award | Soongsil University Spartan SW Education Center 22 Nov 2025
15th Soongsil Capstone Design Competition - Encouragement Award | Soongsil University 1 Oct 2025
2025 College of IT Software Competition - Gold Prize | Soongsil University College of IT 18 Aug 2025
Engineer Information Processing | National Technical Qualification, Ministry of Science and ICT 11 Sep 2026
OPic English - IH (Intermediate High) | Tested 5 Oct 2025 | Valid through 4 Oct 2027